

How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection

Syeda Anshrah Gillani^{a,1,*}, Mirza Samad Ahmed Baig^{b,1,*}, Shahid Munir Shah^c, Abdul Akbar Khan^d, Muhammad Omer Khan^e, Asher Ali^b, Hamzah Siddiqui^b

^a*Heidelberg University, Heidelberg, Germany*

^b*Fandaqah, Al Khobar, Saudi Arabia*

^c*Shaheed Zulfiqar Ali Bhutto Institute of Science and Technology, Karachi, Pakistan*

^d*Argaam Investment, Riyadh, Saudi Arabia*

^e*Fortanixor, Dubai, United Arab Emirates*

Abstract

A quantum-kernel advantage that survives both false-discovery-rate and family-wise correction can be produced entirely by the tuning of the classical baseline. We show this on network intrusion detection, where quantum machine learning routinely reports near-perfect accuracy. Benchmarking hybrid variational circuits and quantum-kernel SVMs against five tuned classical baselines on four NIDS datasets under one leakage-controlled protocol, with a quantum-attribution audit of parameter-matched controls, we find that tuned classical models match or exceed the quantum models on aggregate detection on every dataset, yet a quantum kernel significantly out-ranks its classical surrogate. With exact kernels on identical data, the gap is set by the classical bandwidth grid: under NSL-KDD’s shift to 17 unseen attack types, in-distribution cross-validation cannot rank bandwidths, so a narrow grid yields a spurious ROC-AUC advantage of 0.167 while a wide

*Corresponding authors.

Email addresses: syeda.gilani@stud.uni-heidelberg.de (Syeda Anshrah Gillani), Mirza@fandaqah.com (Mirza Samad Ahmed Baig), dr.shahid@ghr.szabist.edu.pk (Shahid Munir Shah), Akbar.khan@Argaam.com (Abdul Akbar Khan), Omer.khan@Fortanixor.com (Muhammad Omer Khan), CT0@fandaqah.com (Asher Ali), Hamzah@fandaqah.com (Hamzah Siddiqui)

¹Syeda Anshrah Gillani and Mirza Samad Ahmed Baig are the core contributors and contributed equally to this work.

grid leaves 0.007, with the classical kernel ahead at 1% false positives. On UNSW-NB15, which has no unseen attack types, standard selection is reliable and classical kernels win outright, and holding whole attack families out of its training data recreates the failure on demand. Leave-attack-types-out cross-validation repairs selection on NSL-KDD but not under that induced shift, so no training-only rule is a general remedy. The geometric difference of Huang et al. places the IQP kernel within 1.4 to 2.4 of a classical kernel on the circuit’s own phases on all four datasets. On three IBM Heron processors the projected kernel’s classifier stays within 0.01 ROC-AUC of its exact simulation, and the four-qubit hybrid’s one surviving result, higher detection than a tuned Random Forest at 1% false positives, reproduces on every device. Significance is not attribution. Code, seeds, splits and raw device measurements are released.

Keywords: quantum machine learning, network intrusion detection, hybrid quantum-classical neural networks, quantum support vector machine, NISQ, reproducible benchmark, class imbalance, model calibration

1. Introduction

Network intrusion detection systems (NIDS) are a frontline defence against an expanding and increasingly automated threat landscape. Machine learning has become the dominant paradigm for flow- and packet-level intrusion detection, and the recent emergence of quantum machine learning (QML) has prompted a fast-growing body of work applying parameterised quantum circuits, quantum-kernel methods, and hybrid quantum-classical neural networks to the NIDS task. Headline results are striking: accuracies and F1 scores in the 97 to 99.9% range are reported across many datasets and quantum architectures.

These numbers invite a question the field has not answered convincingly: **how quantum is the reported advantage?** There are three reasons for scepticism, each grounded in rigorous recent work. First, the strongest quantum results are typically obtained on heavily dimensionality-reduced inputs, frequently one feature per qubit after principal component analysis (PCA), and on binary tasks evaluated under random cross-validation, a regime that systematically over-states real-world detection. Second, when honest, well-tuned classical baselines (Random Forests, gradient-boosted trees, modern deep networks) are placed on the *same* footing, they repeatedly match or beat the quantum models on tabular IDS data: the Meta-Quantum Ensemble study of Bhatnagar et al. [1], despite assembling the most rigorous

evaluation in the field, explicitly concedes that classical ensembles remain superior. Third, and most pointedly, Bellante et al. [2] demonstrate that the apparent advantage of PCA-based quantum IDS is *explainable by regularisation, not by quantum effects*. A reviewer of any new QML-IDS paper will, and should, raise this point.

We take this scepticism as the starting point rather than something to argue away. A paper whose central claim is “our quantum model achieves higher accuracy” is both scientifically fragile and strategically weak, because three Q1 papers already supply the counter-argument. The community’s actual deficit is not another accuracy record; it is a **fair, reproducible, and honest evaluation methodology**, together with a principled way to **attribute** any measured difference to its true cause. This paper supplies both.

We construct a single, leakage-controlled benchmarking protocol and apply it to the two quantum families that dominate the literature, namely hybrid variational quantum circuits (VQCs) and quantum-kernel support vector machines (QSVMs), against a slate of honestly tuned classical baselines, across four standard datasets. We honour each dataset’s intended evaluation semantics: for NSL-KDD we use the official KDDTrain+/KDDTest+ split, whose test set deliberately over-represents attack behaviour under-seen in training, so that what we measure is generalisation to novel attacks rather than memorisation. We then go beyond benchmarking with a **quantum-attribution audit**: for every quantum model we construct a parameter-matched classical control sharing the identical classical front-end, and we run a regularisation sweep that tests the Bellante et al. hypothesis directly. Finally, because operational IDS must run at a low false-positive operating point and real quantum devices are noisy, we characterise how detection and calibration behave under simulated NISQ noise. Last, we dissect the audit’s strongest positive at the level of the kernels themselves, and show that it was produced by the tuning of the classical baseline under a distribution shift that standard cross-validation cannot see, rather than by the quantum state.

1.1. Contributions

1. **A fair, reproducible, multi-dataset QML-IDS benchmark.** We evaluate hybrid VQCs and QSVMs against five honestly-tuned classical baselines (Random Forest, XGBoost, RBF-SVM, a tuned MLP, a 1D-CNN) across four standard datasets (NSL-KDD, UNSW-NB15, CICIDS2017, NF-ToN-IoT-v2) on the binary detection task, under one

protocol with strict leakage control, honouring official splits. All code, seeds, and splits are released.

2. **An equal-budget fairness design.** We report two explicitly-labelled feature views, a *full* view (classical models on all features) and a *same-budget* view (every model on the identical low-dimensional input the quantum models receive), eliminating the silent feature-budget asymmetry that confounds prior work.
3. **A quantum-attribution audit.** Parameter-matched classical controls and a regularisation sweep decompose any observed gain into its quantum and classical components, a direct constructive test of the Bellante et al. critique in the practical NISQ regime.
4. **A kernel-level dissection of the audit’s strongest positive.** Placing quantum and classical kernels on identical data with exact Gram matrices and equal tuning, we show that a quantum-kernel advantage surviving both FDR and Holm correction is produced by the classical baseline’s bandwidth search, because in-distribution cross-validation cannot see the novel-attack shift of NSL-KDD’s official split. On UNSW-NB15, which lacks that shift, classical kernels win outright.
5. **An induced-shift test, the limits of shift-aware selection, and a label-free dequantisation test.** Holding whole attack families out of UNSW-NB15’s training data reproduces the selection failure on demand. Leave-attack-types-out cross-validation repairs it on NSL-KDD, where the unseen attacks are variants of seen families, but not under the induced shift, so we recommend reporting kernel comparisons under several selection rules with the oracle bound. The geometric difference of Huang et al., computed on all four datasets, shows the IQP kernel to be reproducible by a classical kernel on its own circuit phases; the projected kernel proves robust to finite-shot measurement and, on three IBM Heron processors, to real device noise, and the four-qubit hybrid’s low-false-positive result, the audit’s one surviving positive, reproduces on the same three devices.
6. **Operationally meaningful evaluation under noise.** Beyond accuracy/F1 we report detection rate and false-positive rate together, low-FPR operating points, and calibration, and characterise degradation under depolarising, amplitude-damping, phase-damping, and readout-error channels.
7. **Statistical rigour and honest framing.** Every result is a mean over three to five seeds with confidence intervals, and we report the per-class seed budget explicitly (Section 5.2); quantum-vs-best-classical

comparisons are paired over the seeds the two models share and use McNemar’s test, paired t /Wilcoxon tests, and a paired bootstrap with effect sizes. The benchmark’s value does not depend on the quantum models winning.

2. Related Work

We organise the related work into five strands and summarise representative studies in Table 1.

Foundational and breadth-oriented QML-IDS.. Quantum intrusion detection was established by Kalinin and Krundyshev [3], who applied a quantum SVM and quantum convolutional network to a custom IoT stream dataset ($\sim 98\%$ accuracy); Gong et al. [4] produced the most prominent Q1 variational result (97.21% precision, VQCNN, KDD/NSL-KDD). The strongest *comparative* line is Abreu et al.: QML-IDS [5] and QuantumNetSec [6] span VQC, QSVM, and QCNN across three to four datasets, but with under-tuned classical baselines and no attribution of where gains originate.

Methods landscape.. The dominant primitive is the parameterised quantum circuit used as a variational classifier or hybrid head; recurring concerns are trainability (barren plateaus) and encoding choice, with amplitude embedding reported as unstable relative to angle embedding [7]. Quantum-kernel SVMs [8] form the second family, scaling poorly in sample count, hence the reliance on subsampled, PCA-reduced data. Newer directions include quantum GNNs (Q-AGNN [9]) and meta-learning ensembles [1].

Rigorous and sceptical evaluations.. A small set of studies interrogates whether quantum helps. Bhatnagar et al.’s Meta-Quantum Ensemble [1] sets the field’s evaluation standard (AUPRC, TPR at low FPR, Brier, ECE, five noise channels) yet concludes classical ensembles remain superior. Bellante et al. [2], in *Computers & Security*, show the observed advantage is explainable by regularisation rather than quantum effects. These works define the bar we must clear: not a higher accuracy, but a fair and *attributable* comparison.

Quantum kernels under scrutiny.. Outside intrusion detection, a growing literature asks the same question of quantum kernels. Schnabel and Roth [10] benchmark fidelity and projected quantum kernels over 64 datasets and more than 20,000 models, and find that classical hyperparameters, notably regularisation and bandwidth, matter more than the number of qubits

Table 1: Representative QML-IDS studies and where this work differs (“sim” = simulator).

Paper	Method	Dataset(s)	Honest baseline?	Attribution?
Kalinin & Krundyshev [3]	QSVM, QCNN	Custom IoT stream	No	No
Gong et al. [4]	VQCNN	KDD / NSL-KDD	Partial	No
Abreu et al. [5, 6]	VQC / QSVM / QCNN	UNSW, CICIDS, CIC-IoT	Under-tuned	No
Bhatnagar et al. [1]	QSVM, QNN, RF	CICIDS, TON_IoT	Yes (cl. wins)	No
Bellante et al. [2]	FT PCA-QML	NIDS datasets	Yes	FT/PCA only
This work	VQC, QSVM, controls	4 datasets, official splits	Yes (5 tuned)	Yes (NISQ)

or layers, with no substantial difference between fidelity and projected kernels. Flórez-Ablan et al. [11] show that optimally bandwidth-tuned quantum kernels closely resemble RBF kernels and so hold no quantum advantage, and Shaydulin and Wild [12] establish the importance of the bandwidth itself. Kakavand et al. [13] find no significant advantage for quantum-kernel SVMs across nine tabular datasets, with hardware validation. Abdulsalam and Ahmad [14] compare fidelity quantum kernels with RBF, linear, polynomial and sigmoid kernels, in both classical and quantum SVMs, across eleven tabular classification and regression datasets, and Innan et al. [15] train the quantum kernel variationally to improve QSVM accuracy. Huang et al. [16] supply a label-free criterion, the geometric difference, for when a quantum kernel can outperform a classical one at all. To our knowledge none of these studies examines the novel-attack shift between training and test data that defines NIDS evaluation. Within intrusion detection, Cotrupi and Callahan [17] run quantum kernels on hardware for four attack types, reporting an average accuracy of 85%. We extend this line to that shift, and show how it interacts with the tuning of the classical baseline.

Datasets and evaluation practice. Most studies use one or two datasets (NSL-KDD/KDD99 the dated-but-default choice); genuine multi-dataset and cross-dataset evaluation is rare, and imbalance-aware metrics, calibration, significance testing, and honest baselines are almost universally missing, with Bhatnagar et al. [1] the principal exception.

We differentiate on three axes. Against Abreu et al. we retain breadth

but add tuned baselines, an equal-budget view, operating-point and calibration metrics, significance testing, and reproducibility. Against Bhatnagar et al. we add the causal *decomposition* their paper concedes is missing. Against Bellante et al., the sharpest existing critique of PCA-based quantum IDS, we extend the regularisation argument from their fault-tolerant, PCA-specific case study to the practical NISQ hybrid-VQC and QSVM models the community deploys, across a full multi-dataset benchmark, and we do so constructively: we specify exactly what evidence a genuine advantage would require and report whether, where, and to what extent it appears.

3. Threat Model and Background

3.1. Threat model

We consider flow- and connection-record-based detection: a model classifies each per-connection feature vector as benign or malicious. Three operational constraints shape the evaluation: a strong *cost asymmetry* (false positives flood analysts, so a detector must hold up at a low-FPR operating point); *distribution shift* between training and deployment, including unseen attack signatures (why we honour the NSL-KDD official split); and *class imbalance*. Adaptive adversarial evasion is out of scope.

3.2. Background on quantum machine learning

A QML model operates on n qubits whose joint state is a unit vector in a 2^n -dimensional complex space; a noiseless simulation tracks this 2^n -entry vector, while a noisy simulation tracks a $2^n \times 2^n$ density matrix, whose quadratic cost caps our noisy runs at twelve qubits. Classical flow features must be *encoded* into the circuit (we ablate angle, amplitude, IQP/ZZ, and data-re-uploading encodings); a *variational* ansatz of trainable gates processes the state; and measured Pauli-Z expectations are mapped to class logits by a classical layer (Figure 1). An alternative is to use the circuit only as a fixed feature map and let a classical SVM learn the boundary: the *fidelity kernel* $K(x, x') = |\langle \phi(x') | \phi(x) \rangle|^2$ costs $O(N^2)$ circuit evaluations, while the *projected kernel* computes per-qubit expectations and applies a classical RBF, scaling as $O(N)$.

The link to our central question is direct: in every hybrid pipeline a *classical* front-end compresses high-dimensional records to the per-qubit budget and a *classical* read-out turns measurements into predictions, so any end-to-end accuracy blends a quantum contribution with substantial classical processing and implicit regularisation. Disentangling the two is exactly what the attribution audit (Section 4.4) is built to do.

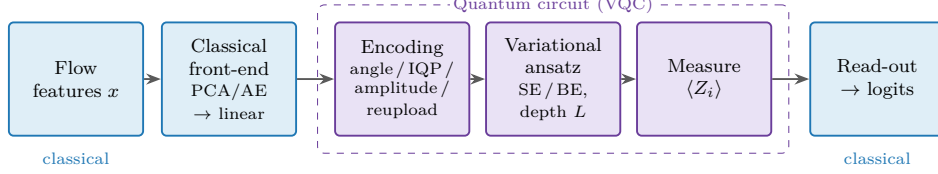


Figure 1: Hybrid quantum-classical classifier. A classical front-end compresses flow features to the per-qubit budget; a variational quantum circuit (encoding + ansatz + measurement) processes the encoded state; a classical read-out produces class logits. The quantum component is sandwiched between classical processing, which is the motivation for the attribution audit.

4. Methodology

4.1. Benchmark protocol and leakage control

We evaluate every model under one protocol designed so no test-distribution information influences training, scaling, reduction, or model selection. Where a dataset ships an official split we honour it: NSL-KDD uses the canonical **KDDTrain+/KDDTest+** split (17 attack signatures appear only in test, so we measure generalisation to novel attacks); UNSW-NB15 uses its official CSVs; CICIDS2017 and NF-ToN-IoT-v2 use a single seeded stratified 70/30 split. All preprocessing is fit on training data only; leaky identifier columns (IP/port four-tuples, flow IDs, timestamps, row indices) are dropped; model selection is by cross-validation *within the training set only*, and the test set is scored once per model per seed.

4.2. Two feature views (equal-budget fairness)

We report a **full view** (classical baselines on all encoded dimensions, the practical ceiling) and a **same-budget view** (every model on the identical PCA-reduced input, $n_{\text{features}} = n_{\text{qubits}} = 8$ in the main configuration). Because one-hot expansion produces many sparse columns, standard-scaling before PCA wastes the budget (eight components retain only $\sim 27\%$ of variance on NSL-KDD); min-max scaling before PCA lifts this to 0.841 (Table 2), so we min-max scale prior to reduction.

4.3. Models

The quantum side is a hybrid VQC family and two QSVMs through a single model factory: a parameter-matched classical MLP control, six hybrid VQCs over the encoding $\times$ ansatz grid (angle, IQP/ZZ, amplitude, reuploading; strongly-entangling [SE] and basic-entangler [BE] ansätze), and

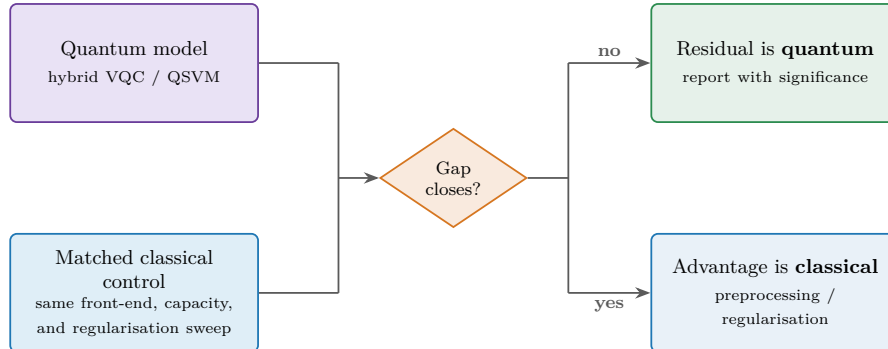


Figure 2: The attribution audit. Each quantum model is compared with a parameter-matched classical control sharing its front-end, plus a regularisation sweep. If a comparably-regularised classical model closes the gap, the apparent advantage is classical; any significant residual is attributed to the quantum circuit.

two QSVMs (IQP fidelity-kernel, angle projected-kernel). The ZZ feature map is built from primitive gates for correct mini-batch broadcasting. The five classical baselines (Random Forest, XGBoost, RBF-SVM, tuned MLP, 1D-CNN) are each hyperparameter-searched (randomised search / grid, 5-fold stratified CV on the training set, optimising F1) and refit per seed.

4.4. The quantum-attribution audit

This is the methodological core and our constructive answer to Bellante et al. [2]. Figure 2 sketches the logic. The audit has three components: (i) *parameter-matched classical controls*: for each hybrid VQC, a classical MLP of matched parameter count sharing the identical front-end, and for each QSVM an RBF-SVM and a *random-feature* kernel approximation; if a matched control equals the quantum model, the “advantage” is not attributable to the quantum component; (ii) a *regularisation sweep* varying classical regularisation (read-out ridge, MLP weight decay, RF depth, SVM bandwidth) to test whether comparable regularisation closes any gap; and (iii) a *gain decomposition* reporting, per dataset and metric, the quantum-minus-control difference with significance.

4.5. Kernel-level diagnostics

The audit above compares models. To explain *why* a quantum kernel does or does not help, we also analyse the kernels themselves. Every quantum kernel studied here is noiseless, so we simulate each feature-map state $|\phi(x)\rangle$ once and form every Gram matrix from the stored statevectors: the

fidelity kernel as $|\Psi_A \Psi_B^\dagger|^2$, and the projected kernel from each qubit’s reduced density matrix. This costs $O(N)$ circuit simulations instead of $O(N^2)$, a shortcut available only in simulation, agrees with the circuits used for the QSVM to within 10^{-15} on every entry, and reproduces the published QSVM test metrics to four decimal places.

Fair surrogates.. We compare quantum and classical kernels on the identical first 2,000 NSL-KDD training rows, with exact Gram matrices and the same kernel SVM solver throughout. The bandwidth γ and $C \in \{0.1, 1, 10, 100\}$ are selected on the training rows only, with the same grid for quantum and classical kernels, so neither side receives more tuning than the other. We use two bandwidth grids, because the choice matters: a narrow one of seven values from 0.003 to 3, and a wide one of ten values from 0.001 to 30, both logarithmically spaced. Selection is either standard 5-fold stratified cross-validation or the shift-aware variant described in Section 7.2.

Induced shift.. To test whether novel attack types cause the selection failure rather than merely accompany it, we repeat the kernel comparison on UNSW-NB15 after removing whole attack categories from the training sample while leaving them in the test sample, with two different sets of removed categories (Section 7.4). As a sensitivity check on the shift-aware rule we also report a second fold construction, which splits benign rows evenly across folds and assigns attack types largest-first to the fold holding the fewest attack rows, so that every fold contains both classes.

Geometric difference and model complexity.. Following Huang et al. [16], for kernels normalised to $\text{Tr } K = N$ we compute

$$g(K_C \| K_Q) = \sqrt{\left\| \sqrt{K_Q} \sqrt{K_C} (K_C + \lambda I)^{-2} \sqrt{K_C} \sqrt{K_Q} \right\|_\infty},$$

which tends to $\sqrt{\left\| \sqrt{K_Q} K_C^{-1} \sqrt{K_Q} \right\|_\infty}$ as $\lambda \rightarrow 0$. A small g certifies that the classical kernel can learn any function the quantum kernel can learn from comparable data, for *any* labelling of the inputs, whereas a prediction advantage requires g to approach $\sqrt{N}$. We take the minimum of g over each classical kernel family and bandwidth, at $N = 800$ and $\lambda = 10^{-2}$. Rank-deficient kernels, such as the linear kernel on eight features, are excluded: their regularised inverse vanishes on the null space, which makes g spuriously small. We also report the label-dependent model complexity $s_K = \left\| \sqrt{K} (K + \lambda I)^{-1} y \right\|^2$ with $y \in \{\pm 1\}$, which tends to $y^\top K^{-1} y$ as $\lambda \rightarrow 0$.

Concentration.. We measure the mean and variance of the off-diagonal kernel entries as the qubit count grows from 2 to 12. Exponential concentration [18] would drive this variance to zero exponentially in the number of qubits.

Finite shots.. Each Pauli expectation μ in the projected kernel is replaced by the mean of M outcomes in $\{\pm 1\}$ drawn with $P(+1) = (1 + \mu)/2$, which is what hardware returns from M shots in each of the three measurement bases, and the SVM is retrained on the resulting noisy features.

Hardware execution.. Because the projected kernel needs only single-qubit expectations, the feature map can be run on a real device as one parameterised eight-qubit circuit with 24 observables, executed once per sample: linear in the number of samples, unlike the $O(N^2)$ fidelity kernel. We build the identical IQP circuit in Qiskit (its statevector expectations agree with the PennyLane feature map to 10^{-15}), transpile it to the device’s native gate set (depth 119, 28 two-qubit gates on the heavy-hex coupling map), and evaluate $\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle$ with the Estimator primitive at 128 shots per measurement basis, with the primitive’s default readout-error mitigation and no other error mitigation. The 200 training rows are the first 200 of the NSL-KDD training file and the 500 test rows are a seeded, class-balanced sample of the official test set. All 700 samples are submitted as a single job. We compare the measured features and the resulting Gram matrix with their exact statevector values by Pearson correlation, and train and test the projected-kernel SVM on the measured features alone. As a control we first run the same pipeline on a noisy local simulation of the device built from its calibration data.

The same devices also run the variational model behind the audit’s one surviving positive, the four-qubit hybrid of Table 4. Its trained circuit (angle encoding, three strongly entangling layers) is rebuilt in Qiskit with the trained weights bound as constants and the four input angles as circuit parameters (statevector expectations agree with PennyLane to 10^{-15}), transpiled to the heavy-hex coupling map (depth 81 to 92 and 21 to 30 two-qubit gates, depending on the layout chosen for the device’s current calibration), and its four $\langle Z_i \rangle$ are measured at 128 shots for a seeded, class-balanced sample of 2,000 test rows, as one job per device. The measured expectations pass through the trained classical head unchanged, so only the quantum layer is replaced. The model is retrained from the released configuration and seed before the run; the retrained model reproduces the published test scores to 2×10^{-8} . We compare it with the tuned Random Forest of Table 4

Table 2: Datasets (binary task; PCA-EVR@8 = variance retained by 8 principal components).

Dataset	Split	Raw dim	Classes (bin / multi)	Train / test	EVR@8
NSL-KDD [20]	Official	122	2 / 40	125,973 / 22,544	0.841
UNSW-NB15 [21]	Official	194	2 / 10	175,341 / 82,332	0.870
CICIDS2017 [22]	Strat. 70/30	76	2 / 3	729,491 / 312,639	0.924
NF-ToN-IoT-v2 [23]	Strat. 70/30	39	2 / 10	140,000 / 60,000	0.974

on the same 2,000 rows by paired bootstrap.

Classical phase kernels.. Under angle encoding every qubit remains in a product state, so its Pauli features are exactly $(\sin x_i, 0, \cos x_i)$ and the “quantum” projected kernel is an RBF kernel on classical trigonometric features. We therefore also test RBF kernels on $\varphi(x) = (\sin x_i, \cos x_i)_i$, and on φ augmented with the phases that the ZZ feature map [19] writes into the state, namely sin and cos of $2x_i$ and of $2(\pi - x_i)(\pi - x_{i+1})$. This isolates whether an advantage requires the quantum state itself, or only the phases it encodes.

4.6. Class imbalance and noise

The NSL-KDD binary task is only mildly imbalanced, so we default to cost-sensitive learning (balanced class weights, `scale_pos_weight` for XGBoost, weighted cross-entropy) and treat SMOTE (training partition only) as an ablation. For noise robustness we sweep four channels (depolarising, amplitude damping, phase damping, and bit-flip readout) at strengths $p \in \{0, 0.001, 0.005, 0.01, 0.05\}$ on PennyLane’s `default.mixed` density-matrix backend, capped at twelve qubits.

5. Experimental Setup

5.1. Datasets

We evaluate on four standard NIDS datasets (Table 2). The **main benchmark is the binary task**. CICIDS2017 here is the three-day *MachineLearningCVE* subset (~ 1.04 M flows); NF-ToN-IoT-v2 is stratified-downsampled to 200,000 rows (from 16.9M); both are standard, configurable scopes we disclose plainly.

5.2. Metrics and significance

We report threshold metrics (accuracy, precision, recall, F1, detection rate DR, false-positive rate FPR) and ranking/calibration metrics (ROC-AUC, AUPRC, TPR@0.1%FPR, TPR@1%FPR, Brier, ECE), matching and extending the Meta-Quantum-Ensemble suite [1]. We state the seed budget exactly, because it is uneven across model classes. The tuned classical baselines are run over five seeds {42, 43, 44, 45, 46} on NSL-KDD and over three seeds {42, 43, 44} on UNSW-NB15, CICIDS2017 and NF-ToN-IoT-v2; the quantum models and the attribution-audit controls over three seeds {42, 43, 44}; and the noise-robustness sweep over two seeds {42, 43}, with all RNGs pinned per seed and t -based 95% confidence intervals reported throughout. Every quantum-versus-classical comparison is therefore paired over the three seeds {42, 43, 44} that all models share, and uses McNemar’s test (shared test set), paired t /Wilcoxon tests with Cohen’s d (over seeds), and a paired bootstrap (2,000 resamples) for operating-point metrics. Because the attribution audit evaluates many variant $\times$ metric comparisons, we control the false-discovery rate across this family with the Benjamini-Hochberg (BH) procedure and report BH-adjusted q -values alongside raw p -values; we treat a result as robust only if it survives FDR control.

5.3. Backend efficiency and reproducibility

At QML-IDS qubit counts, quantum simulation (not GPU throughput) dominates runtime, and we report a **backend crossover**: the CPU `lightning.qubit` simulator is markedly faster than the GPU `lightning.gpu` simulator (Section 6.5). Each run writes a self-describing JSON record (config, per-seed metrics, tuned hyperparameters, reference-seed predictions, environment snapshot with source checksums). All experiments run on a single laptop (NVIDIA RTX 3050 Ti, 4GB, under WSL2); the main runs use eight qubits, with a ceiling of sixteen (noiseless) / twelve (noisy). Code, seeds, and splits are released.

6. Results

The overall result, qualified below: *honestly-tuned classical models match or exceed the quantum models on aggregate detection on every dataset, but a small hybrid VQC retains a statistically significant edge at the strict low-FPR operating point on the distribution-shifted NSL-KDD task.*

Table 3: Headline benchmark. F1 (mean over seeds): best classical per view vs. the leading 8-qubit quantum model per dataset (same-budget input). The final three columns give that quantum model’s AUPRC, ROC-AUC, and TPR@1%FPR. Stronger quantum ablation variants (e.g. the IQP-QSVM and the 4-qubit hybrid on NSL-KDD, F1 0.752 and 0.759) are analysed in Section 6.2.

Dataset	Classical F1		Quantum (8 qubits, same-budget)			
	full	same-budget	F1	AUPRC	ROC-AUC	TPR@1%FPR
NSL-KDD	0.803	0.782	0.730	0.911	0.865	0.411
UNSW-NB15	0.912	0.891	0.836	0.942	0.924	0.577
CICIDS2017	1.000	1.000	0.968	0.992	0.998	0.975
NF-ToN-IoT-v2	0.996	0.991	0.949	0.965	0.969	0.195

Best classical model per cell. Full view: SVM (NSL-KDD), MLP (UNSW), RF (CICIDS, NF-ToN-IoT-v2); same-budget: XGBoost (NSL-KDD), MLP (UNSW), RF (CICIDS, NF-ToN-IoT-v2).

6.1. Headline benchmark

Table 3 reports F1 under both views. Under the *same-budget* view (the apples-to-apples comparison), the best classical baseline beats the leading 8-qubit quantum model on every dataset: NSL-KDD 0.782 vs. 0.730; UNSW-NB15 0.891 vs. 0.836; CICIDS2017 1.000 vs. 0.968; NF-ToN-IoT-v2 0.991 vs. 0.949. On NSL-KDD the strongest quantum *ablation* variants narrow this gap substantially, since the IQP-encoded QSVM and the 4-qubit hybrid reach F1 of 0.752 and 0.759 (Section 6.2), but tuned classical still leads on F1, and retains a clearer margin on AUPRC and ROC-AUC, so the aggregate ordering is unchanged. Two qualifications matter. First, **CICIDS2017 and NF-ToN-IoT-v2 are near-saturated** (every tuned classical model scores $F1 \geq 0.97$), so they have weak discriminative power and the quantum models’ high absolute scores there should be read as “this task is easy.” Second, the **CV-to-test gap** confirms why literature numbers are inflated: on NSL-KDD, models scoring $F1 \approx 0.98$ under within-training cross-validation collapse to $F1 \approx 0.78$ on the official test split, a consistent ≈ 0.20 drop across all models and both views. Figure 3 compares the best classical and best quantum F1 per dataset; the qubit-count ablation is shown later in Figure 4.

6.2. Attribution audit

The audit (Table 4) decomposes the difference on NSL-KDD, the discriminative dataset. A parameter-matched classical MLP sharing the quantum front-end attains AUPRC 0.944 and the tuned Random Forest attains F1 0.764, both matching or exceeding every hybrid VQC on aggregate metrics. For the twelve-qubit hybrid the classical controls win significantly (F1 -0.039 , $p = 0.021$; AUPRC -0.036 , $p = 0.035$; TPR@1%FPR -0.121 , $p = 0.026$). **Once a classical model is granted the same front-end and comparable capacity/regularisation, the aggregate “quantum advantage” disappears**, a confirmation of the Bellante hypothesis on this dataset. Two exceptions survive Benjamini-Hochberg FDR control across the audit’s comparison family, however, and they are the most informative results in the audit. First, the quantum-kernel SVM significantly out-performs the random-feature (Nyström / RBF-sampler) kernel: on NSL-KDD the IQP-encoded QSVM beats the random-feature control on AUPRC ($\Delta = +0.075$, raw $p = 0.001$, BH $q = 0.011$), ROC-AUC ($\Delta = +0.134$, $p = 0.002$, $q = 0.018$), and TPR@1%FPR ($\Delta = +0.152$, $p = 0.017$, $q = 0.047$). Taken at face value this would isolate a genuinely quantum contribution, since the random-feature kernel is meant as the classical surrogate for the quantum feature map. Section 7 shows that it does not: the control differed from the QSVM in four further ways, and once those are removed the gap is set by the classical bandwidth search, while a classical kernel on the circuit’s own phases reproduces the quantum kernel. Second, the *four-qubit* hybrid beats the tuned Random Forest at the low-FPR operating point (TPR@1%FPR $+0.050$, $p = 0.005$, BH $q = 0.030$; Section 6.3). The remaining quantum differences are either ties or significant *losses* to the classical controls (notably the twelve-qubit hybrid: F1 -0.039 , $p = 0.021$; AUPRC -0.036 , $p = 0.035$), consistent with overfitting at width rather than a quantum benefit. Within the audit, then, classical wins on aggregate and two FDR-robust quantum differences persist: the quantum kernel over the random-feature control, which Section 7 traces to the control, and the small hybrid at the strict operating point.

6.3. Operating-point and calibration analysis

The operating-point metrics hold one of the audit’s two FDR-robust quantum differences; the other, the quantum-kernel-vs-random-feature-kernel comparison, is analysed in Section 7. On NSL-KDD the four-qubit hybrid VQC achieves TPR@1%FPR = 0.517, exceeding the best classical model (Random Forest, 0.467) by 0.050, statistically significant ($p = 0.005$,

Table 4: Quantum-attribution audit (NSL-KDD): leading quantum variants vs. best matched control. Δ = quantum – control (positive favours quantum, except ECE). q is the Benjamini-Hochberg FDR-adjusted value across the audit family; **bold** rows survive FDR control ($q < 0.05$). “Hyb.-Angle qn ” is the angle-encoded hybrid VQC on n qubits; rf-kern is the random-feature kernel and MLP-m the parameter-matched MLP.

Variant	Metric	Quantum	Best control	Δ	Q better?	p	q (BH)
QSVM-IQP	AUPRC	0.957	0.882 (rf-kern)	+0.075	yes	0.001	0.011
QSVM-IQP	ROC-AUC	0.945	0.811 (rf-kern)	+0.134	yes	0.002	0.018
QSVM-IQP	TPR@1%FPR	0.470	0.318 (rf-kern)	+0.152	yes	0.017	0.047
Hyb.-Angle q4	TPR@1%FPR	0.517	0.467 (RF)	+0.050	yes	0.005	0.030
Hyb.-Angle q4	TPR@0.1%FPR	0.407	0.272 (MLP-m)	+0.135	yes	0.428	0.62
Hyb.-Angle q4	F1	0.759	0.764 (RF)	−0.004	tie	0.776	0.84
Hyb.-Angle q12	F1	0.724	0.764 (RF)	−0.039	no	0.021	0.10
Hyb.-Angle q12	TPR@1%FPR	0.346	0.467 (RF)	−0.121	no	0.026	0.11

paired bootstrap). At 0.1%FPR it reaches 0.407 against 0.272, a difference that is not significant ($p = 0.43$). At the strict 1%-alarm budget a defender would actually impose, the small quantum model catches a larger fraction of attacks on novel-attack-heavy traffic. Calibration is comparable (ECE ≈ 0.18 to 0.22 for hybrids vs. 0.193 for Random Forest), the four-qubit hybrid marginally better. This is the pattern a genuine, if narrow, quantum signal would produce, though it survives FDR and not Holm correction (Section 6.5).

6.4. Noise robustness

Sweeping four NISQ channels on the six-qubit hybrid shows **graceful degradation, and in the mild regime none at all**. Relative to the noiseless run (ROC-AUC 0.846, TPR@1%FPR 0.422, ECE 0.191), depolarising, amplitude-damping, and phase-damping noise at $p \in \{0.01, 0.05\}$ leave low-FPR detection essentially unchanged (0.421 to 0.430) and slightly *raise* ROC-AUC (to between 0.878 and 0.897), consistent with mild noise acting as a regulariser on the shifted task. Only bit-flip (readout) noise materially hurts (TPR@1%FPR 0.379 at $p = 0.05$). No channel causes catastrophic collapse: for this small-circuit regime, readout error is the one to mitigate.

6.5. Significance and efficiency

We control the false-discovery rate across the audit family (108 direction-aware comparisons on NSL-KDD) with Benjamini-Hochberg; 13 quantum-favouring and 27 classical-favouring differences survive at $q < 0.05$, so on

Table 5: Backend efficiency. CPU (`lightning.qubit`) vs. GPU (`lightning.gpu`) training-step time at fixed work.

Qubits	CPU (s)	GPU (s)	GPU/CPU
4	1.12	8.89	8.0×
8	2.08	28.0	13.5×
12	3.94	54.3	13.8×

balance the corrected evidence still favours classical on aggregate. On aggregate F1 the classical controls significantly beat quantum (twelve-qubit hybrid vs. RF, raw $p = 0.021$). The two quantum differences that survive FDR control are the quantum-kernel-vs-random-feature-kernel comparison (AUPRC $q = 0.011$, ROC-AUC $q = 0.018$) and the four-qubit hybrid at the low-FPR point ($q = 0.030$). We note honestly that the latter survives FDR (BH) but not the more conservative family-wise Holm correction across the full surface, under which only the quantum-kernel-vs-surrogate results remain significant. Statistical robustness is not attribution, however: Section 7 shows that the Holm-robust kernel result is produced by its classical control. The backend-crossover finding (Table 5) is unambiguous: the **CPU simulator is 8 to 14× faster than the GPU** at QML-IDS qubit counts, the gap growing with width because the per-circuit work is too small to amortise GPU kernel-launch and transfer overhead. The projected quantum kernel ($O(N)$) is tractable at test scale whereas the fidelity kernel ($O(N^2)$) is not.

7. Kernel-Level Analysis: How Quantum Is the Kernel Advantage?

The audit’s most robust positive was the quantum-kernel SVM out-ranking its random-feature surrogate (Section 6.2), the one comparison to survive both FDR and family-wise Holm correction. That control, however, differed from the QSVM in four ways besides the feature map: it was trained on 1,200 rather than 2,000 rows, it approximated its kernel with 512 Nyström components, it used a linear rather than a kernel SVM, and its bandwidth $\gamma = 1$ was applied untuned to raw features on a different scale. This section removes each difference in turn, examines how the result depends on hyperparameter selection, tests it on a second official split, and asks whether what remains requires the quantum state, and finally runs the kernel on real devices. The diagnostics are those of Section 4.5.

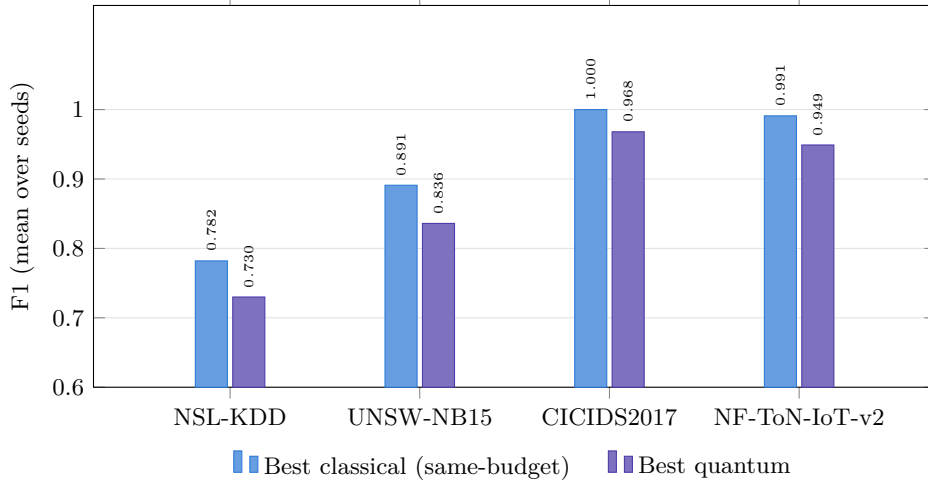


Figure 3: Best classical (same-budget) vs. best quantum F1 per dataset (mean over seeds). Tuned classical models lead on every dataset; the gap is largest on the discriminative datasets (NSL-KDD, UNSW-NB15) and smallest where the task is near-saturated (CICIDS2017, NF-ToN-IoT-v2, where every classical model exceeds 0.97). The vertical axis is truncated at 0.6.

7.1. The surrogate gap is set by the classical bandwidth search

Table 6 places every kernel on the identical 2,000 NSL-KDD training rows, with exact Gram matrices and one kernel SVM solver. Removing the first three confounds, with the bandwidth still fixed at $\gamma = 1$, narrows the published ROC-AUC gap from 0.127 to 0.085. Hyperparameter selection then decides the outcome. With the bandwidth searched over $\gamma \leq 3$, cross-validation picks $\gamma = 0.3$ for the raw-feature RBF kernel, whose test ROC-AUC falls to 0.783: an apparent quantum advantage of 0.167 (95% CI [0.161, 0.173]). With the same cross-validation over $\gamma \leq 30$ it picks $\gamma = 30$, and the RBF kernel reaches 0.940. The quantum kernel then leads by only 0.007 in ROC-AUC ([0.006, 0.009]) and *trails* at the operating point, where the classical kernel detects 0.455 of attacks at 1% false positives against 0.442 ($\Delta = -0.013$, $[-0.020, -0.005]$, $p = 0.002$). The advantage the audit reported is therefore not a property of the kernels. It appears or disappears with the width of the classical baseline’s search grid, a special case of the bandwidth sensitivity documented for quantum kernels by Shaydulin and Wild [12], here acting on the *classical* side of the comparison. Schnabel and Roth [10] likewise find bandwidth and regularisation to dominate quantum-kernel performance; what is new here is that a distribution shift makes the

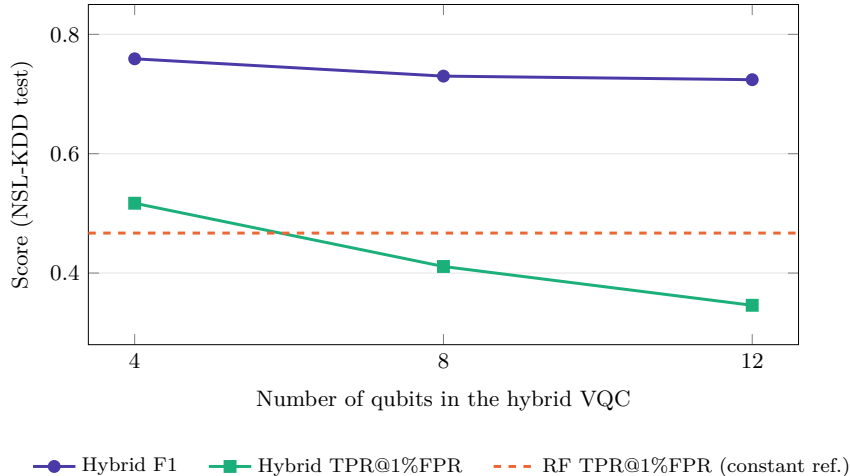


Figure 4: Qubit-count ablation of the hybrid VQC on NSL-KDD. Both F1 and the low-FPR detection rate *decline* with width, so more qubits do not help on the shifted task. Crucially, only the small four-qubit circuit exceeds the best classical baseline’s TPR@1%FPR (Random Forest, dashed reference at 0.467); the advantage vanishes by twelve qubits, consistent with an implicit-regularisation mechanism rather than raw expressivity.

classical baseline’s choice of them unobservable to cross-validation.

7.2. Cross-validation cannot see the novel-attack shift

Figure 5 shows why the grid matters so much. Within the training distribution every kernel scores between 0.95 and 0.99 at every bandwidth for $C = 1$, so the cross-validation surface is nearly flat and its maximum is close to arbitrary. On the official test set, whose 17 attack types never occur in training, the raw-feature RBF kernel is U-shaped in γ : strong at very wide and very narrow bandwidths (up to 0.945), and weakest at intermediate ones (0.740 at worst across the grid), which is exactly where a grid capped at $\gamma = 3$ leads cross-validation. Across the 40 settings of the grid the RBF kernel’s standard CV score is uncorrelated with its test ROC-AUC ($r = -0.16$).

This motivates *shift-aware model selection*: leave-attack-types-out cross-validation, in which each attack type lives in exactly one fold, so every validation fold contains attack types its training fold never saw. It reproduces the condition of the official split using training data alone; attack types are assigned to folds by GroupKFold, and the three of five folds in which both classes occur are used. Its score correlates with test ROC-AUC at

Table 6: Kernels on identical data (first 2,000 NSL-KDD training rows; official test set of 22,544 rows containing 17 attack types absent from training). Exact Gram matrices and one kernel SVM solver throughout. *Score* is the selection criterion’s own value: in-distribution 5-fold CV AUC, or leave-attack-types-out CV AUC for shift-aware rows. The phase kernel is an RBF kernel on $(\sin x_i, \cos x_i)$ plus the ZZ feature map’s phases, with no quantum state. Fixed rows use $\gamma = 1$, $C = 1$. Generated from the result files.

Kernel	Feat.	Selection	Score	Test AUC	AUPRC	TPR@1%FPR
<i>Quantum</i>						
IQP projected, published	24	fixed	n/a	0.945	0.957	0.470
IQP projected	24	CV, $\gamma \leq 3$	0.993	0.950	0.959	0.444
IQP projected	24	CV, $\gamma \leq 30$	0.994	0.948	0.957	0.442
IQP projected	24	shift-aware CV	0.941	0.945	0.962	0.499
IQP fidelity	256	CV, C only	0.992	0.947	0.955	0.458
<i>Classical, no quantum state</i>						
Random features, 1,200 rows, published	512	fixed	n/a	0.818	0.885	0.352
Random features, 2,000 rows	512	fixed	n/a	0.830	0.878	0.053
RBF, raw features	8	fixed	n/a	0.861	0.892	0.045
RBF, raw features	8	CV, $\gamma \leq 3$	0.993	0.783	0.864	0.078
RBF, raw features	8	CV, $\gamma \leq 30$	0.994	0.940	0.953	0.455
RBF, raw features	8	shift-aware CV	0.948	0.939	0.951	0.466
Trig $(\sin x_i, \cos x_i)$ only	16	CV, $\gamma \leq 3$	0.993	0.811	0.878	0.077
Phase kernel	46	CV, $\gamma \leq 3$	0.995	0.941	0.952	0.446
Phase kernel	46	CV, $\gamma \leq 30$	0.995	0.938	0.946	0.441
Phase kernel	46	shift-aware CV	0.951	0.947	0.959	0.469

$r = +0.79$ for the RBF kernel, and it selects robust settings for every kernel family, with test ROC-AUC between 0.939 and 0.947 (Table 6). Under this rule the quantum projected kernel also reaches the highest operating-point detection we observe, $\text{TPR@1\%FPR} = 0.499$, above the RBF kernel’s 0.466 ($\Delta = +0.033$, $[0.021, 0.067]$, $p < 0.001$). We do not read this as a quantum advantage, because its sign depends on the selection rule: under standard CV the same comparison favours the classical kernel by 0.013. Having now evaluated three selection rules, reporting only the one on which quantum wins would be precisely the researcher degree of freedom this paper warns against.

7.3. A second official split

If the collapse of standard CV is caused by novel attack types, it should vanish on a dataset whose test set has none. UNSW-NB15 is the only other benchmark here with an official split, and every one of its nine attack

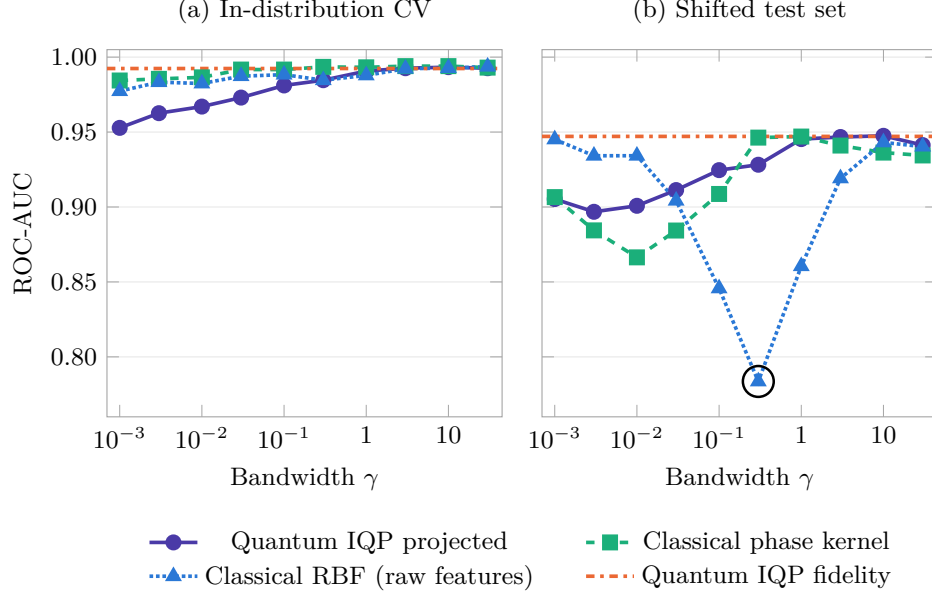


Figure 5: Kernel performance across the RBF bandwidth on NSL-KDD ($C = 1$, first 2,000 training rows). (a) Within the training distribution every kernel scores 0.95 to 0.99 at every bandwidth, so cross-validation cannot tell good settings from bad ones. (b) On the official test set, whose 17 attack types never occur in training, the raw-feature RBF kernel collapses at intermediate bandwidths while recovering at both extremes. Standard CV over a grid capped at $\gamma = 3$ selects $\gamma = 0.3$ ($C = 100$), test ROC-AUC 0.783, at the circled dip; with the grid extended to $\gamma = 30$ it selects $\gamma = 30$ ($C = 1$), test ROC-AUC 0.940. The classical phase kernel uses the ZZ feature map’s phases with no quantum state; the IQP fidelity kernel has no bandwidth and is drawn as a constant. On this dataset the quantum projected kernel never falls below 0.893 across the full grid, against 0.740 for the raw-feature RBF; on UNSW-NB15 the order reverses (Section 7.3).

categories occurs in both training and test. We repeat the protocol on a seeded, stratified sample of 2,000 training rows and 20,000 test rows; the training file is sorted by class, so its first rows are all benign and cannot be used. Table 7 confirms the prediction: on UNSW-NB15 standard CV tracks test performance for every kernel ($r = +0.82$ to $+0.83$). The result for the quantum kernel, however, reverses. Selected by the same rule, it is significantly worse than both classical kernels on every metric: the classical phase kernel of Section 7.5 reaches test ROC-AUC 0.956 against 0.930 ($\Delta = -0.026$, $[-0.028, -0.024]$) and detects 0.751 of attacks at 1% false positives against 0.457 ($\Delta = -0.294$, $[-0.318, -0.263]$), and even the raw-feature RBF kernel is ahead ($\Delta\text{TPR} = -0.057$, $[-0.087, -0.022]$, $p = 0.005$). The

Table 7: Kernels on the two datasets with an official train/test split, each selected by standard 5-fold CV over the full grid ($\gamma \leq 30$, $C \leq 100$). *Worst* is the lowest test ROC-AUC over all 40 settings; r is the correlation, across those settings, between the CV score and the test ROC-AUC. NSL-KDD’s test set contains 17 attack types absent from training; UNSW-NB15’s contains none. Generated from the result files.

Dataset	Kernel	Test AUC	AUPRC	TPR@1%FPR	Worst	r
NSL-KDD	Quantum IQP projected	0.948	0.957	0.442	0.893	+0.83
	Phase kernel (classical)	0.938	0.946	0.441	0.818	+0.24
	RBF, raw features (classical)	0.940	0.953	0.455	0.740	−0.16
UNSW-NB15	Quantum IQP projected	0.930	0.943	0.457	0.697	+0.82
	Phase kernel (classical)	0.956	0.968	0.751	0.787	+0.83
	RBF, raw features (classical)	0.938	0.951	0.514	0.789	+0.83

one property that had favoured the quantum kernel on NSL-KDD, fewer poor hyperparameter settings (worst test ROC-AUC 0.893 against 0.740), does not survive either: on UNSW-NB15 its worst setting is the lowest of the three (0.697 against 0.787 and 0.789). We therefore do not claim it as a property of quantum kernels. Shift-aware CV on UNSW-NB15, whose official split has no novel attacks for it to anticipate, is examined with the induced-shift experiment in Section 7.4.

7.4. Inducing the shift

The contrast between the two datasets is consistent with the novel-attack explanation but does not test it, because the datasets differ in more than their splits. We therefore induce the shift on UNSW-NB15 itself: whole attack categories are removed from the 2,000-row training sample and left in the 20,000-row test sample, and the protocol of Table 7 is repeated. Two holdouts guard against a lucky choice. Holdout A removes Exploits, Fuzzers and Reconnaissance, which then make up 25% of the test rows; holdout B removes Generic, DoS, Analysis and Backdoor, 29% of the test rows. Table 8 gives the result.

Unseen attack types are necessary for the failure but not sufficient. Under holdout A standard CV still ranks the grid correctly for every kernel ($r = +0.85$ to $+0.99$) and its choices lie within 0.009 ROC-AUC of the best setting in the grid. Under holdout B the NSL-KDD picture returns. Every kernel loses between 0.06 and 0.16 ROC-AUC at its best setting, the

correlation between CV score and test ROC-AUC falls to +0.62 to +0.69, and the loss from selection is unequal across kernels: the CV-selected raw-feature RBF kernel sits 0.108 below its oracle (0.794 against 0.902), the quantum kernel 0.028 below (0.761 against 0.789). A comparison made at the CV-selected settings therefore understates the classical kernel’s lead by 0.08 ROC-AUC, and at 1% false positives it reverses the order outright: the quantum kernel detects 0.302 of attacks against 0.263 for the RBF kernel, whose best setting in the grid detects 0.465. This is the mechanism of Section 7.2 reproduced on demand, on a dataset on which it was absent, by changing nothing but which attack types the training sample contains. What separates the two holdouts is the bandwidth landscape: under holdout B the RBF kernel’s test ROC-AUC peaks sharply at $\gamma = 0.3$ (0.902, against 0.794 at $\gamma = 0.1$ and 0.864 at $\gamma = 1$) while its CV score is flat across those settings (0.966 to 0.967), so the selection criterion cannot resolve the one setting that generalises to the unseen categories.

Shift-aware selection does not repair holdout B. Leave-attack-types-out CV, which restored selection on NSL-KDD, correlates with test ROC-AUC at $r = +0.43$ for the RBF kernel here and chooses a setting worse than standard CV (0.751). Its folds can only hold out attack types the training sample contains (Exploits, Fuzzers, Reconnaissance, Shellcode, Worms), and a shift towards those types is not the shift towards Generic and DoS that the test set presents. On NSL-KDD the seventeen unseen types are variants within the same four attack families as the seen ones, which is why holding out types reproduces the official split there. Nor is the result an artefact of how the folds are built. The construction used throughout assigns attack types to folds with GroupKFold, which leaves three usable folds on NSL-KDD and one on UNSW-NB15; an alternative that splits the benign rows evenly and assigns attack types largest-first to the emptiest fold makes all five folds usable on every dataset, but gives $r = +0.54$ instead of +0.79 for the RBF kernel on NSL-KDD, $r = -0.38$ under holdout B, and on the official UNSW-NB15 split, which has no unseen types, it selects an over-smooth bandwidth for the RBF kernel (test ROC-AUC 0.859 against 0.938 under standard CV). Shift-aware selection is a choice of target rather than a free improvement: it buys robustness to the kinds of novelty the training data can imitate, at a cost when there is no novelty to be robust to, and it cannot anticipate novelty of a kind the training data has never seen. The general lesson is procedural. A kernel comparison under distribution shift should be reported under more than one selection rule, with the oracle bound alongside, so that a reader can see how much of a claimed difference is selection.

Table 8: Inducing novel-attack shift on UNSW-NB15. Whole attack categories are removed from the 2,000-row training sample and kept in the 20,000-row test sample: holdout A removes Exploits, Fuzzers, Recon. (25% of test rows); holdout B removes Generic, DoS, Analysis, Backdoor (29% of test rows). Each kernel is selected by standard 5-fold CV over the full grid ($\gamma \leq 30$, $C \leq 100$); r is the correlation across the 40 settings between the CV score and the test ROC-AUC, and *oracle* the best test ROC-AUC in the grid. The last two columns give TPR@1%FPR at the CV-selected setting and the best in the grid. Generated from the result files.

Unseen types	Kernel	r	Test ROC-AUC		TPR@1%FPR	
			CV pick	Oracle	CV pick	Best
None (0%)	Quantum IQP projected	+0.82	0.930	0.948	0.457	0.561
	Phase kernel (classical)	+0.83	0.956	0.957	0.751	0.751
	RBF, raw features (classical)	+0.83	0.938	0.957	0.514	0.661
Holdout A (25%)	Quantum IQP projected	+0.99	0.936	0.940	0.739	0.744
	Phase kernel (classical)	+0.85	0.938	0.947	0.714	0.764
	RBF, raw features (classical)	+0.92	0.940	0.946	0.708	0.779
Holdout B (29%)	Quantum IQP projected	+0.69	0.761	0.789	0.302	0.344
	Phase kernel (classical)	+0.62	0.844	0.861	0.413	0.487
	RBF, raw features (classical)	+0.69	0.794	0.902	0.263	0.465

7.5. Geometry: the IQP kernel is close to its own classical phases

Table 9 gives the geometric difference on all four datasets, which requires no labels and no hyperparameter selection. The angle-encoded fidelity kernel is reproduced by an RBF kernel to $g = 1.06$ to 1.07 on every dataset, so by the bound of Huang et al. [16] it can hold no prediction advantage over that classical kernel for any labelling. This is expected: under angle encoding every qubit remains in a product state, and the kernel factorises into a product of cosines. The IQP kernels, which use entangling ZZ gates, are genuinely far from standard kernels: against every RBF bandwidth and against plain trigonometric features, g ranges from 4.9 to 12.2. Flórez-Ablan et al. [11] show that optimally rescaling the encoding drives quantum kernels towards RBF kernels; our IQP kernels use the unscaled encoding of the original pipeline, and consistently with that result they sit far from RBF. The same IQP kernels, however, lie within $g = 1.4$ to 2.4 of a purely classical kernel built on the phases the ZZ feature map writes [19], on all four datasets, and their label-dependent model complexities agree: on NSL-KDD the quantum projected kernel has $s_K = 235$ and the classical phase kernel $s_K = 236$, against 481 for the best raw-feature RBF. The empirical results are consistent with this geometry. On NSL-KDD the phase kernel matches the quantum kernel at the operating point under a narrow grid (0.446 against 0.444, $p = 0.89$) and under standard CV over the full grid

Table 9: Geometric difference $g(K_C \| K_Q)$, minimised over classical bandwidth ($N = 800$, $\lambda = 10^{-2}$, $\sqrt{N} \approx 28.3$). A small g means the classical kernel can learn anything the quantum kernel can. The last column gives the angle-encoded fidelity kernel against its best RBF. Rank-deficient kernels are excluded. Generated from the result files.

Dataset	IQP kernel	Classical family, vs IQP kernel			Angle fidelity
		RBF raw	Trig	Phase kernel	vs RBF
NSL-KDD	projected	8.41	8.33	2.27	1.07
	fidelity	4.91	4.88	1.77	
UNSW-NB15	projected	12.21	12.10	2.44	1.07
	fidelity	9.52	9.34	1.50	
CICIDS2017	projected	8.65	8.48	1.95	1.06
	fidelity	7.04	6.86	1.54	
NF-ToN-IoT-v2	projected	9.25	9.07	2.23	1.07
	fidelity	7.84	7.64	1.39	

(0.441 against 0.442, $p = 0.90$), and on UNSW-NB15 it outperforms it. Whatever the IQP kernel contributes lives in the phases of its circuit, not in the quantum state, and a classical kernel that uses those phases obtains it at a small fraction of the cost.

7.6. Concentration and finite shots

Kernel values concentrate as width grows, but mildly at these sizes. From 2 to 12 qubits the mean off-diagonal IQP fidelity falls from 0.32 to 0.047, yet its variance falls only from 0.099 to 0.032 and flattens between 8 and 10 qubits, far from the exponential regime of Thanasilp et al. [18]. The kernels’ test ROC-AUC peaks at 8 to 10 qubits (0.944 to 0.946 for the projected kernel) and dips only to 0.927 at 12. Concentration therefore does not explain the decline of the variational hybrid with width (Figure 4), which points instead to the trainability and capacity of the variational model.

The projected kernel is robust to measurement noise (Table 10). With 16 shots in each of three measurement bases, 48 shots per sample in total, test ROC-AUC falls from 0.945 to 0.941, and AUPRC and TPR@1%FPR do not move beyond their run-to-run spread; by 64 shots the result is indistinguishable from the exact kernel. This follows from the construction: the projected kernel needs only single-qubit expectations, each estimated with variance at most $1/M$, which the RBF then smooths, whereas a fidelity kernel must resolve small global overlaps. The study models sampling noise only; Section 7.7 tests the same kernel, with its gate and readout errors, on real devices.

Table 10: Finite-shot robustness of the published IQP projected QSVM on NSL-KDD. M shots per basis in each of the X, Y and Z settings; mean and standard deviation over 5 sampling seeds. Generated from the result files.

Shots / basis	Shots / sample	ROC-AUC	AUPRC	TPR@1%FPR
16	48	0.941 ± 0.002	0.956 ± 0.002	0.486 ± 0.011
64	192	0.945 ± 0.002	0.960 ± 0.001	0.489 ± 0.009
256	768	0.943 ± 0.004	0.957 ± 0.002	0.473 ± 0.003
1,024	3,072	0.943 ± 0.002	0.956 ± 0.001	0.469 ± 0.002
4,096	12,288	0.947 ± 0.001	0.957 ± 0.001	0.470 ± 0.001
exact	∞	0.945	0.957	0.470

7.7. On real hardware

Table 11 reports the projected kernel executed on three IBM Heron processors, `ibm_fez`, `ibm_kingston` and `ibm_marrakesh`, on the same 200 training and 500 test samples, each as a single job of 89 to 91 QPU-seconds (Section 4.5). The measured Pauli features correlate with their exact values at $r = 0.92$ to 0.97 (RMSE 0.10 to 0.15), and the Gram matrices built from them at $r = 0.97$ to 0.98 ; on `ibm_fez` the Gram correlation of 0.977 matches the $r \geq 0.976$ that Kakavand et al. [13] report for the same device. The classifier trained and tested on measured features alone reaches ROC-AUC 0.935, 0.935 and 0.942 against 0.944 for the exact kernel: a deficit of 0.009 on `ibm_fez` (95% CI [0.000, 0.018], $p = 0.049$) and on `ibm_kingston` ([0.002, 0.018], $p = 0.017$), and none on `ibm_marrakesh` ([−0.002, 0.007], $p = 0.35$). AUPRC shows a significant deficit only on `ibm_kingston` (0.011, $p = 0.002$), and at the 1% false-positive operating point no device differs from the exact kernel (measured 0.404 to 0.408 against 0.396, $p \geq 0.62$). The mechanism anticipated in Section 7.6 holds on hardware: per-feature noise well above the shot-noise floor is largely absorbed by the RBF over 24 features.

Two caveats apply. The devices were noisier at the feature level than the calibration-based noise model of the dry run predicted (test RMSE 0.15 on `ibm_fez` against 0.07 in simulation), so noise models of this kind should not be taken as substitutes for device runs. And hardware validity is not advantage: the same measured features are, by Section 7.5, reproducible by a classical kernel on the circuit’s phases. What the device runs establish is that the object we analysed is realisable at current noise levels, at a cost of about 0.13 QPU-seconds per sample.

The device runs also settle the status of the audit’s one surviving positive.

Table 11: Hardware validation of the IQP projected kernel on NSL-KDD (200 training and 500 test samples, the latter class-balanced). Agreement columns are Pearson correlations of the source’s Pauli features and of its Gram matrix (off-diagonal) with the exact statevector values. Metric columns are for the projected-kernel SVM trained and tested on that source’s own features ($\gamma = 1$, $C = 1$). Generated from the result files.

Feature source	Agreement with exact		Projected-kernel SVM		
	Feat. r	Gram r	ROC-AUC	AUPRC	TPR@1%FPR
Exact statevector	1.000	1.000	0.944	0.941	0.396
Noisy simulation (FakeFez, 256 shots)	0.984	0.990	0.941	0.938	0.420
<code>ibm_fez</code> (128 shots)	0.920	0.977	0.935	0.935	0.404
<code>ibm_kingston</code> (128 shots)	0.941	0.971	0.935	0.930	0.404
<code>ibm_marrakesh</code> (128 shots)	0.966	0.984	0.942	0.939	0.408

Section 6.2 left the four-qubit hybrid’s advantage over the tuned Random Forest at 1% false positives as a hypothesis for hardware testing; Table 12 tests it. On 2,000 class-balanced test rows the retrained model’s exact scores give ROC-AUC 0.900 and $\text{TPR@1\%FPR} = 0.496$, against 0.929 and 0.304 for the Random Forest on the same rows, so the pattern of Table 4 is present in this sample: the forest is better on aggregate ($\Delta\text{ROC-AUC} = -0.029$, $[-0.042, -0.017]$) and the hybrid better at the low-false-positive operating point ($\Delta\text{TPR} = +0.192$, $[0.009, 0.296]$, $p = 0.031$). With the four expectations measured on `ibm_fez`, `ibm_kingston` and `ibm_marrakesh` instead of simulated, they correlate with their exact values at $r = 0.97$ to 0.98 (RMSE 0.11 to 0.12, of which shot noise at 128 shots accounts for up to 0.09), the final scores at $r = 0.99$, and the model’s ROC-AUC falls by 0.017 to 0.020 on every device ($p < 0.001$): close to the 0.012 that shot noise alone costs in simulation, with a device contribution below 0.01. The operating-point result is untouched. $\text{TPR@1\%FPR}$ is 0.498, 0.498 and 0.501 on the three devices against 0.496 exact ($p \geq 0.9$), and the margin over the Random Forest on the same rows is +0.194, +0.194 and +0.197 ($p = 0.033$, 0.058 and 0.047). The low-FPR result therefore survives real device noise on three processors, which is more than the simulated audit could say. It remains what it was: a comparison between a variational model and a tree ensemble on one dataset, resting here on 1,000 benign rows, and not an attribution

to the quantum layer, whose four-qubit circuit is well within classical reach. The cost was 86 to 87 QPU-seconds per device for 2,000 samples, about 0.04 QPU-seconds per sample.

Table 12: The 4-qubit hybrid on hardware: NSL-KDD, 2,000 class-balanced test rows. Only the four $\langle Z_i \rangle$ that feed the trained classical head are measured; every other part of the model is the published one. r is the Pearson correlation of the source’s expectations with the exact statevector values. Metric columns are the hybrid’s own on that source; the shot-noise row is the mean of 20 binomial resamplings of the exact expectations. The last column is the paired-bootstrap difference in TPR@1%FPR against the tuned random forest on the same rows (positive favours the hybrid; two-sided p in brackets). Generated from the result files.

Source of $\langle Z_i \rangle$	r	Hybrid QNN			
		ROC-AUC	AUPRC	TPR@1%FPR	Δ TPR vs RF
Exact statevector	1.000	0.900	0.914	0.496	+0.192 [0.03]
Shot noise only (128 shots)		0.888	0.909	0.502	
FakeFez simulation (128 shots)	0.981	0.881	0.905	0.495	
ibm_fez (128 shots)	0.976	0.883	0.906	0.498	+0.194 [0.03]
ibm_kingston (128 shots)	0.972	0.882	0.904	0.498	+0.194 [0.06]
ibm_marrakesh (128 shots)	0.976	0.880	0.904	0.501	+0.197 [0.05]
Tuned random forest, same rows		0.929	0.917	0.304	

7.8. Summary

Once kernels are compared on identical data with equal and adequate tuning, no quantum-kernel advantage survives the choice of selection rule or of dataset. The audit’s strongest positive was produced by its own classical surrogate: first by four confounds, then by a bandwidth grid too narrow for a distribution shift that in-distribution cross-validation cannot see. Three findings of general use come out of the exercise. The selection failure can be induced at will by holding attack families out of training, and no training-only rule repairs it in general: leave-attack-types-out cross-validation restores selection on NSL-KDD, where the unseen attacks are relatives of seen ones, but not under the induced shift, so kernel comparisons under shift should be reported under several rules with the oracle bound. The geometric difference identifies, without labels, when a quantum kernel is classically reproducible, which here it is from its own circuit phases on every dataset. And the objects under analysis run on current hardware: on three IBM Heron processors the projected kernel’s measured Gram matrix correlates at 0.97 to 0.98 with simulation and its classifier stays within 0.01 ROC-AUC

of the exact kernel, and the four-qubit hybrid keeps its low-false-positive margin over the Random Forest, so neither the classical reproducibility nor the one surviving positive is an artefact of simulation.

8. Discussion

How quantum is the advantage? On this benchmark and in the practical NISQ regime, almost none of it. The audit settles the aggregate question: once a classical model is granted the same front-end and comparable regularisation, the aggregate advantage of the hybrid VQC and the QSVM vanishes, and tuned classical baselines significantly outperform the quantum models on F1 and AUPRC. We thus confirm the Bellante et al. [2] hypothesis and extend it from their fault-tolerant, PCA-specific case study to the deployed NISQ models, and we corroborate Bhatnagar et al.’s concession [1] while supplying the decomposition their work lacked.

The kernel-level analysis settles the exception we had expected to be the most robust. The quantum kernel’s advantage over its classical surrogate survived both FDR and family-wise correction, yet it was not quantum: it came from four confounds in the surrogate and then from the width of the classical bandwidth grid (Section 7). This is a cautionary result for the field, not only for this paper. Statistical robustness does not establish attribution, and under the distribution shift that makes NIDS evaluation meaningful, the classical baseline’s hyperparameter search can manufacture or erase an apparent quantum advantage. A QML-IDS claim evaluated on a split with unseen attack types should report how its classical baselines were tuned, over what range, and by which selection rule, and should report the comparison under more than one rule with the oracle bound, since no training-only rule, shift-aware selection included, proved reliable under every shift we induced.

One narrower positive remains from the audit. At the 1%-FPR operating point on NSL-KDD the four-qubit hybrid out-detects the tuned Random Forest ($p = 0.005$, BH $q = 0.030$), although it does not survive the more conservative Holm correction. It appears on the discriminative dataset rather than the saturated ones, and it is strongest for the small circuit and vanishes at twelve qubits, consistent with implicit regularisation rather than expressivity. Our noise sweep used a six-qubit hybrid, so it does not bear on this configuration. On three IBM Heron processors it reproduces with the quantum layer’s expectations measured rather than simulated (Section 7.7), so it is not an artefact of noiseless simulation; but it remains a comparison

between a variational model and a tree ensemble on one dataset, a comparison the kernel-level controls of Section 7 do not address, and we do not regard it as evidence of quantum advantage.

What remains classical. For a practitioner choosing a detector today, the honest recommendation is a tuned Random Forest or XGBoost: they win or tie on every aggregate metric, on every dataset, at a fraction of the cost and with no quantum hardware. Three methodological findings generalise. The ≈ 0.20 F1 CV-to-test gap is a caution for the whole QML-IDS literature; leave-attack-types-out cross-validation addresses the model-selection failure behind it on NSL-KDD, and the induced-shift experiment shows where such rules stop working; and the backend crossover (the GPU is an order of magnitude *slower* at these qubit counts) saves researchers real time.

9. Limitations and Threats to Validity

Hardware. The projected quantum kernel and the four-qubit hybrid’s quantum layer were executed on three IBM Heron processors (Section 7.7); every other quantum result, including the wider hybrids and the noise sweep, is simulated, and the noise sweep models NISQ effects rather than measuring them. The device runs used 200 training and 500 test samples for the kernel and 2,000 test samples for the hybrid, 128 shots per basis, and only the primitive’s default readout mitigation; they realise inference, not training, on hardware, and the fidelity kernel, whose cost is quadratic in the sample count, was not run. The hybrid’s low-false-positive margin on hardware rests on 1,000 benign rows, so its 1% threshold is set by ten scores and its confidence intervals are wide. **Qubit budget.** The 4 GB GPU caps width at roughly sixteen qubits (noiseless) / twelve (noisy, $(2^n)^2$ memory); the main configuration uses eight, so we cannot probe the larger-width regime. **Dimensionality reduction** discards variance (eight PCs retain $\sim 84\%$ on NSL-KDD), so any advantage is conditioned on information-reduced input. **Dataset age/composition.** NSL-KDD is dated (mitigated by three further datasets); the IoT coverage is a single NetFlow dataset; CICIDS2017 and NF-ToN-IoT-v2 use standard subsets, configurable to full corpora. **Trainability** (barren plateaus, gradient variance) is not characterised directly, and QSVMs are trained on subsampled data ($O(N^2)$ fidelity kernel). **Statistical power.** Our seed budget is small and uneven (five seeds for the classical baselines on NSL-KDD, three elsewhere and for all quantum runs, two for the noise sweep), so paired tests over seeds have low power: the two-sided Wilcoxon floor at three paired seeds is $p = 0.25$, which is why the operating-point claims rest on the paired bootstrap over

the fixed test set rather than on seed-level tests. The crossover ratio is hardware/library-dependent. **Kernel analysis.** The kernel-level analysis uses eight qubits, the angle and IQP feature maps, noiseless statevectors, and the two datasets with official splits; the geometric difference covers all four datasets, at $N = 800$. Shift-aware cross-validation was informative on NSL-KDD but not on UNSW-NB15, where one attack category dominates, and it requires attack-type labels that production data may not provide. The NSL-KDD QSVM, as in the original pipeline, trains on the first 2,000 rows of the official training file, so that model has a single realisation rather than three independent seeds. **Scope.** Binary detection under benign distribution shift and imbalance; adversarial evasion and the unsupervised/online/federated settings are out of scope.

10. Conclusion

We set out to replace the field’s recurring “our quantum model reaches 99%” claim with a fair, reproducible, attribution-aware yardstick. Across four datasets under one leakage-controlled protocol, well-tuned classical models match or exceed the quantum models on aggregate detection everywhere, and the audit attributes this to classical preprocessing and regularisation rather than quantum effects, confirming and generalising the Bellante et al. critique. The audit’s strongest positive, a quantum kernel that out-ranked its classical surrogate under both false-discovery-rate and family-wise correction, did not survive a kernel-level analysis. It was produced by the surrogate’s confounds and by the classical bandwidth grid, under a distribution shift that standard cross-validation cannot see, and a classical kernel on the circuit’s own phases reproduces the quantum kernel on all four datasets. The lesson reaches beyond this paper: in QML benchmarks evaluated under distribution shift, significance is not attribution, and the tuning of the classical baseline must be reported as carefully as the quantum model itself. We contribute the benchmark and harness, the kernel-level diagnostics, an induced-shift test, and a shift-aware selection rule together with its limits, and release them so that QML-IDS claims can be stated, and contested, on common, honest ground.

Reproducibility.. Code, configurations, fixed seeds, split scripts, and per-run provenance are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>; the preprocessed datasets will be released on request.

CRedit authorship contribution statement

Syeda Anshrah Gillani and Mirza Samad Ahmed Baig are the core contributors and contributed equally to all aspects of the work. **Syeda Anshrah Gillani:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review & editing), Visualization, Supervision. **Mirza Samad Ahmed Baig:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review & editing), Visualization, Supervision. **Shahid Munir Shah:** Methodology, Validation, Writing (review & editing). **Abdul Akbar Khan:** Validation, Writing (review & editing). **Muhammad Omer Khan:** Validation, Writing (review & editing). **Asher Ali:** Resources, Writing (review & editing). **Hamzah Siddiqui:** Investigation, Validation, Writing (review & editing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used are public benchmarks (NSL-KDD, UNSW-NB15, CICSIDS2017, NF-ToN-IoT-v2) obtainable from their original providers. The code, configurations, fixed seeds, the download and preprocessing scripts that reconstruct our exact splits, and the per-run result records are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>. We do not redistribute the source corpora; the preprocessed datasets will be released on request.

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used AI-assisted tooling for coding assistance. The authors reviewed and edited all content and take full responsibility for the content of the publication.

References

- [1] R. Bhatnagar, N. Innan, A. A. Jothi, M. Shafique, Meta-quantum ensemble framework for robust network intrusion detection, arXiv preprint (2026). `arXiv:2605.28879`.
URL <https://arxiv.org/abs/2605.28879>
- [2] A. Bellante, T. Fioravanti, M. Carminati, S. Zanero, A. Luongo, Evaluating the potential of quantum machine learning in cybersecurity: A case-study on PCA-based intrusion detection systems, *Computers & Security* (2025). `arXiv:2502.11173`.
URL <https://arxiv.org/abs/2502.11173>
- [3] M. Kalinin, V. Krundyshev, Security intrusion detection using quantum machine learning techniques, *Journal of Computer Virology and Hacking Techniques* 19 (1) (2022) 125–136. doi:10.1007/s11416-022-00435-0.
- [4] C. Gong, et al., Network intrusion detection based on variational quantum convolution neural network, *The Journal of Supercomputing* (2024). doi:10.1007/s11227-024-05919-y.
- [5] D. Abreu, C. E. Rothenberg, A. Abelém, QML-IDS: Quantum machine learning intrusion detection system, arXiv preprint (2024). `arXiv:2410.16308`.
URL <https://arxiv.org/abs/2410.16308>
- [6] D. Abreu, C. E. Rothenberg, A. Abelém, QuantumNetSec: Quantum machine learning for network security, *International Journal of Network Management* (2025). doi:10.1002/nem.70018.
- [7] Z. Wang, K.-W. Fok, V. L. L. Thing, Network attack traffic detection with hybrid quantum-enhanced convolution neural network, arXiv preprint (2025). `arXiv:2504.20436`.
URL <https://arxiv.org/abs/2504.20436>
- [8] S. Kumari, S. R. Pokhrel, et al., Modeling wavelet transformed quantum support vector for network intrusion detection, arXiv preprint (2025). `arXiv:2512.01365`.
URL <https://arxiv.org/abs/2512.01365>
- [9] D. Chaudhary, S. Rajasegarar, S. R. Pokhrel, Q-AGNN: Quantum-enhanced attentive graph neural network for intrusion detection, arXiv

preprint (2026). [arXiv:2603.22365](https://arxiv.org/abs/2603.22365).
URL <https://arxiv.org/abs/2603.22365>

- [10] J. Schnabel, M. Roth, Quantum kernel methods under scrutiny: a benchmarking study, *Quantum Machine Intelligence* 7 (2025) 58. doi:10.1007/s42484-025-00273-5.
- [11] R. Flórez-Ablan, M. Roth, J. Schnabel, On the similarity of bandwidth-tuned quantum kernels and classical kernels, *Quantum Science and Technology* 10 (3) (2025) 035051. doi:10.1088/2058-9565/ade7ad.
- [12] R. Shaydulin, S. M. Wild, Importance of kernel bandwidth in quantum machine learning, *Physical Review A* 106 (2022) 042407. doi:10.1103/PhysRevA.106.042407.
- [13] S. Kakavand, C. Strohmeyer, M. Schlotter, Benchmarking quantum kernel support vector machines against classical baselines on tabular data: A rigorous empirical study with hardware validation (2026). [arXiv:2604.18837](https://arxiv.org/abs/2604.18837).
- [14] G. Abdulsalam, I. Ahmad, Comparative investigation of quantum and classical kernel functions applied in support vector machine algorithms, *Quantum Information Processing* 24 (4) (2025) 109. doi:10.1007/s11128-025-04728-3.
- [15] N. Innan, M. A. Z. Khan, B. Panda, M. Bennai, Enhancing quantum support vector machines through variational kernel training, *Quantum Information Processing* 22 (10) (2023) 374. doi:10.1007/s11128-023-04138-3.
- [16] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, J. R. McClean, Power of data in quantum machine learning, *Nature Communications* 12 (2021) 2631. doi:10.1038/s41467-021-22539-9.
- [17] M. L. Cotrupi, B. R. Callahan, Towards practical quantum kernels for network intrusion detection, in: *Proceedings of the AAAI Symposium Series*, Vol. 7, 2025, pp. 339–342. doi:10.1609/aaais.v7i1.36903.
- [18] S. Thanasilp, S. Wang, M. Cerezo, Z. Holmes, Exponential concentration in quantum kernel methods, *Nature Communications* 15 (2024) 5200. doi:10.1038/s41467-024-49287-w.

- [19] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, *Nature* 567 (2019) 209–212. doi:10.1038/s41586-019-0980-2.
- [20] M. Tavallae, E. Bagheri, W. Lu, A. A. Ghorbani, A detailed analysis of the KDD CUP 99 data set, in: *IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*, 2009. doi:10.1109/CISDA.2009.5356528.
- [21] N. Moustafa, J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection systems, in: *Military Communications and Information Systems Conference (MilCIS)*, 2015. doi:10.1109/MilCIS.2015.7348942.
- [22] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: *International Conference on Information Systems Security and Privacy (ICISSP)*, 2018. doi:10.5220/0006639801080116.
- [23] M. Sarhan, S. Layeghy, M. Portmann, Towards a standard feature set for network intrusion detection system datasets, *Mobile Networks and Applications* 27 (2022) 357–370. doi:10.1007/s11036-021-01843-0.